

(a)(i)

What is meant by a constant acceleration of 0.32 m s^{-2} ?

The velocity increases by 0.32 m s^{-1} each second.

 *Revision tip:* Constant acceleration means the **velocity changes at the same rate** every second — like adding the same amount to your speed every second.

(a)(ii)

Calculate the distance travelled by the train in the first 25 s.

Use:

$$s = ut + \frac{1}{2}at^2$$
$$s = 0 + \frac{1}{2} \times 0.32 \times 25^2 = 0.16 \times 625 = 100\text{ m}$$

 *Revision tip:*

Use the **suvat** equations for motion with constant acceleration. If **u = 0**, the distance is entirely from acceleration.

(b)(i)

Calculate the frequency heard by a stationary observer as the train approaches.

Use the **Doppler effect** for approaching source:

$$f_o = f_s \left(\frac{v}{v - v_s} \right)$$
$$f_o = 340 \left(\frac{340}{340 - 50} \right) = 340 \times \frac{340}{290} = 340 \times 1.1724 = 398\text{ Hz}$$

 *Revision tip:*

As the **source moves toward the observer**, the **waves bunch up**, increasing the frequency heard. This is the **Doppler effect**.

(b)(ii)

Explain why the frequency heard by an observer on the platform is lower after the train passes.

Once the train passes, the wavefronts are more spaced out behind the source, meaning **fewer wavefronts per second** reach the observer.

 *Revision tip:* Remember that **wavelength increases** behind a receding source, so **frequency decreases** for the observer.